

Lecture 16 Hypergraph container & maximal Δ -free graphs.

Hypergraph container method

{ Babai-Morris-Sarnakij
Saxton-Thomason.

Roughly speaking, given a hypergraph (w/ 'nice' edge distribution)
clustering phenomena of its independent sets.

$\exists$ few containers s.t. $\forall$ indep $I \subseteq$ one of container



Thm 1.1 (Hypergraph container thm Informal)

Let $\mathcal{H}$ be a uniform hypergraph. If its edges are 'evenly' distributed (bdd codes), then $\exists$ a family of containers $\mathcal{C} \subseteq 2^{V(\mathcal{H})}$ s.t.

- 1) $\forall$ indep set I in $\mathcal{H}$, $\exists C \in \mathcal{C}$ s.t. $I \subseteq C$;
- 2) # containers $C \in \mathcal{C}$ is small;
- 3) each container $C \in \mathcal{C}$ is almost independent, $e(\mathcal{H}[C])$ is small.

Phrasing problems in terms of ind set in hypergraphs

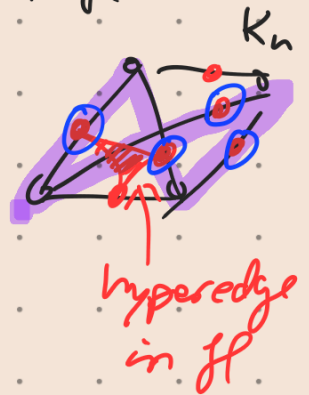
Consider $\mathcal{G}$ = all Δ -free graphs
on vertex set $[n]$ (labeled)

Construct a
• 3-unif hypergraph $\mathcal{H} = (V, E)$ where

- vertices in $\mathcal{H}$ are edges in K_n : $V = E(K_n)$

- each triangle in $K_n \longrightarrow$ an edge in $\mathcal{H}$

• independent set I in $\mathcal{H} \iff$ triangle-free
graph $\in \mathcal{G}$



e.g. $\# \Delta$ -free graphs
on $[n] = \#$ indep sets
in $\mathcal{H}$

Thm 1.2 (containers for triangle-free graphs)

For $\forall \delta > 0$, $\exists c = c(\delta) > 0$ s.t. $\exists$ a family of graphs

$\mathcal{F} \subseteq 2^{\binom{[n]}{2}}$ satisfying the following:

1) $\forall \Delta$ -free graph G on $[n]$, $\exists F \in \mathcal{F}$ s.t.

$G \subseteq F$;

2) there are at most $2^{c \cdot n^{3/2} \log n}$ graphs in $\mathcal{F}$;
 ($2^{o(n^2)}$ many)

3) each $F \in \mathcal{F}$ is almost Δ -free: $\leq \sqrt{n^3} \Delta_s$.

Recall: Δ removal lem: $\# \Delta_s \text{ in } F = o(n^3)$

$$\Rightarrow F = A \cup B \leftarrow e(B) = o(n^2)$$

$\uparrow$ Δ -free graph $\Rightarrow$ Markel $e(A) \leq \frac{n^2}{4}$

4) $\forall F \in \mathcal{F}, \quad e(F) \leq (\frac{1}{4} + o(1))n^2$

Counting maximal Δ -free graphs.

Q: How many Δ -free graphs on $[n]$?
 (i.e. size of $\mathcal{G} = \text{all } \Delta \text{ free graphs on } [n]$)

Lower bd: $K_{n/2, n/2} \xrightarrow{\text{take all possible subgraphs}} \geq 2^{n^2/4} \text{ graph}$

$\bigcirc \bigcirc \bigcirc$

each is Δ -free.

exponential error

[Erdős-Kleitman-Rothschild 76] $(\frac{1}{4} + o(1))n^2$



• $\# \Delta$ -free graphs on $[n] = 2$

• typical structure: almost all Δ -free graphs are bipartite

Warm up: (Reprove Erdős-Kleitman-Rothschild bound)

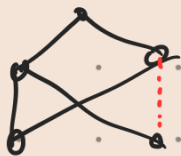
$$\# \Delta\text{-free graphs on } [n] \stackrel{(1)}{\leq} \sum_{F \in \mathcal{F}} 2^{e(F)/4} \leq \sum_{F \in \mathcal{F}} 2^{(\frac{1}{4} + o(1))n^2}$$

Containers for Δ -free graphs

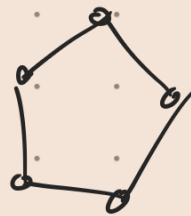
$$= |\mathcal{F}| \cdot 2^{(\frac{1}{4} + o(1))n^2} \stackrel{(2)}{\leq} 2^{o(n^2)} \cdot 2^{(\frac{1}{4} + o(1))n^2} = 2^{(\frac{1}{4} + o(1))n^2}$$

Def. A Δ -free graph G is maximal Δ -free if adding any non-edge to G creates a Δ .

Ex



No maximal



maximal Δ -free

Problem (Erdős) Estimate $\#$ maximal Δ -free graphs.

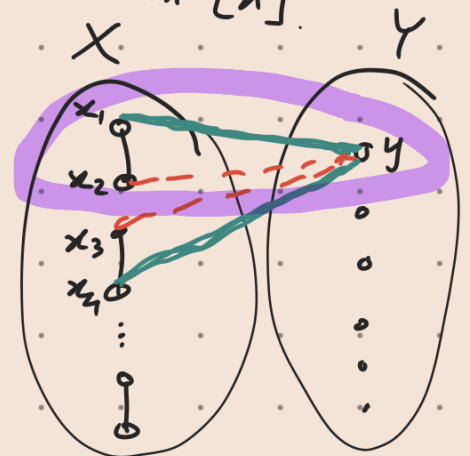
Lower bd: $\geq 2^{n^2/8}$ many maximal Δ -free graphs on $[n]$.

• Take $|X| = |Y| = n/2$

X : perfect matching

Y : indep.

• For each pair of $(\text{edge}, \frac{v_x}{x})$



add exactly one edge $e(x) \rightarrow y$ $|Y|$
 $\# \text{choices} = 2^{\# \text{pairs}} = 2^{\frac{n}{4} \cdot \frac{n}{2}} = 2^{n^2/8}$

all Δ -free

- since no more edges can be added to $[x, Y]$
 each such graph can be extended to a distinct
 max. Δ -free graph (distinguished by the cut
 $[x, Y]$)

$$2^{n^2/8} \leq \leq 2^{n^2/4 + o(n^2)}$$

Thm (Balogh - Petrickova 14)

$$\frac{n^2}{8} + o(n^2)$$

$$\# \text{ maximal } \Delta\text{-free graphs on } [n] = 2$$

Remark : Typical structure of maximal Δ -free is
 also known now. (Balogh - Liu - Petrickova - Sharifzadeh) (15)

Tool Thm Hajtler-Tuza

$$\forall \Delta\text{-free } n\text{-vx graph } G, \# \text{ maximal indep. sets in } G \leq 2^{n/2}$$

• Tight : $G = \downarrow \downarrow \dots \downarrow$

• If not Δ -free, Moon-Moser (60s) $\text{mis} \leq 3^{n/3} \triangle \triangle \dots \triangle$

Pf $\left(\# \text{ max. } \Delta\text{-free} = 2^{n^2/8 + o(n^2)} \right)$

$\# \text{ max. } \Delta\text{-free} \leq \sum_{\substack{F \in \mathcal{F} \\ \uparrow \\ \text{containers for } \Delta\text{-free}}} \# \text{ max. } \Delta\text{-free subg. in } F$

As $|\mathcal{F}| \leq 2^{o(n^2)}$, suffices to show

$\forall F \in \mathcal{F}, \# \text{ max. } \Delta\text{-free in } F \leq 2^{n^2/8 + o(n^2)}$

Fix arbitrary $F \in \mathcal{F}$.

$\Delta\text{-removal lem} \Rightarrow F = A \cup B$

- A is $\Delta\text{-free}$

- $B: e(B) = o(n^2)$

Build max. $\Delta\text{-free}$ G_A in $F = A \cup B$,

$G = G_A \cup G_B$ with $G_A \subseteq A$ and $G_B \subseteq B$

Step 1 Choose intersection w/ B

$\Delta\text{-free } G_B$

$\# \text{ choices} \leq 2^{e(B)} = 2^{o(n^2)}$

Step 2 Extend $G_B \subseteq B$ in A to a max. $\Delta\text{-free}$

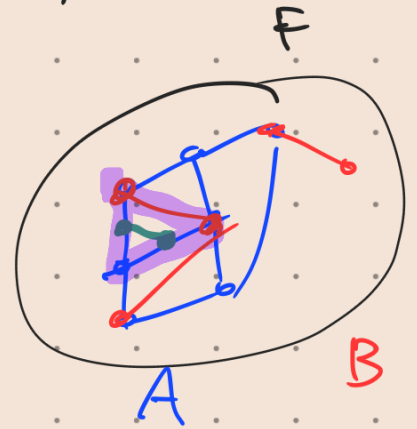
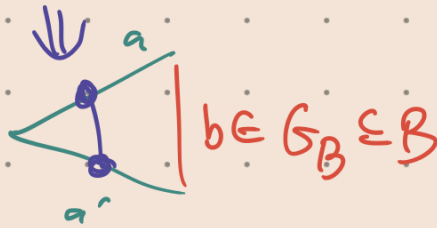
NTS $\# \text{ extensions} \leq 2^{n^2/8 + o(n^2)}$ for each fixed choices

$\# \text{ extensions} \leq \# \text{ max. indep. sets in } \mathcal{P} \text{ of } G_B$

Consider an auxiliary graph Γ , where

$$V(\Gamma) = E(A)$$

$$\begin{aligned} a \sim a' &\iff \exists b \in G_B \\ aa' \in E(\Gamma) &\iff aa'b = \Delta \end{aligned}$$



Obs: Each extensib. of G_B in $A \rightsquigarrow$
maximal indep set is Γ

(As $G_B \cup G_A$ must be maximal Δ -free)

$$\Rightarrow \boxed{\# \text{extensions} \leq \# \text{max. indep. sets in } \Gamma}$$

$$G \subseteq F = A \cup B$$

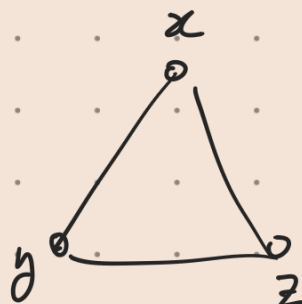
Claim Γ is Δ -free.

Claim $\Rightarrow$ thm Huiter-Tuza:

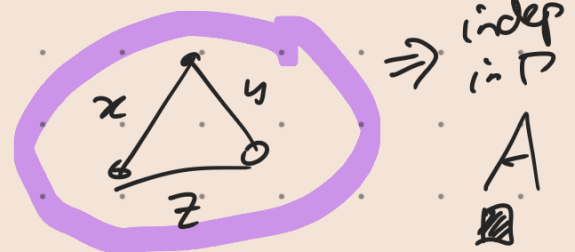
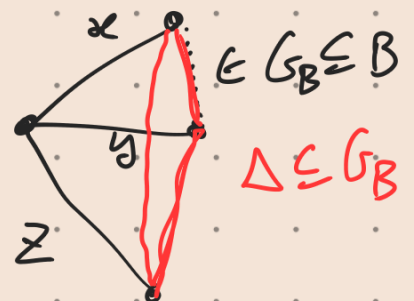
$$\# \text{max. indep sets in } \Gamma \leq 2^{|\Gamma|/2} \leq 2^{n^2/8}$$

$$\begin{aligned} \# \text{ubs in } \Gamma &= |\Gamma| \\ &= e(A) \\ &\leq n^2/4 \end{aligned}$$

Pf Claim



Δxyz in Γ
 $x, y, z \in E(A)$



OPEN # max. K_4 -free graphs?

Conj 2 $\frac{1}{2} \text{ex}(n, K_4) \rightarrow o(n^2)$